

Marketing Analytics

Data Analysis Report — **SQL Queries** · **Python Sentiment Analysis** · **Power BI Dashboards**





The Problem

An online retail business faces three interconnected challenges undermining marketing ROI:

Declining Engagement

Fewer interactions with site content and marketing campaigns.

Low Conversion

Fewer site visitors completing purchases despite high traffic.

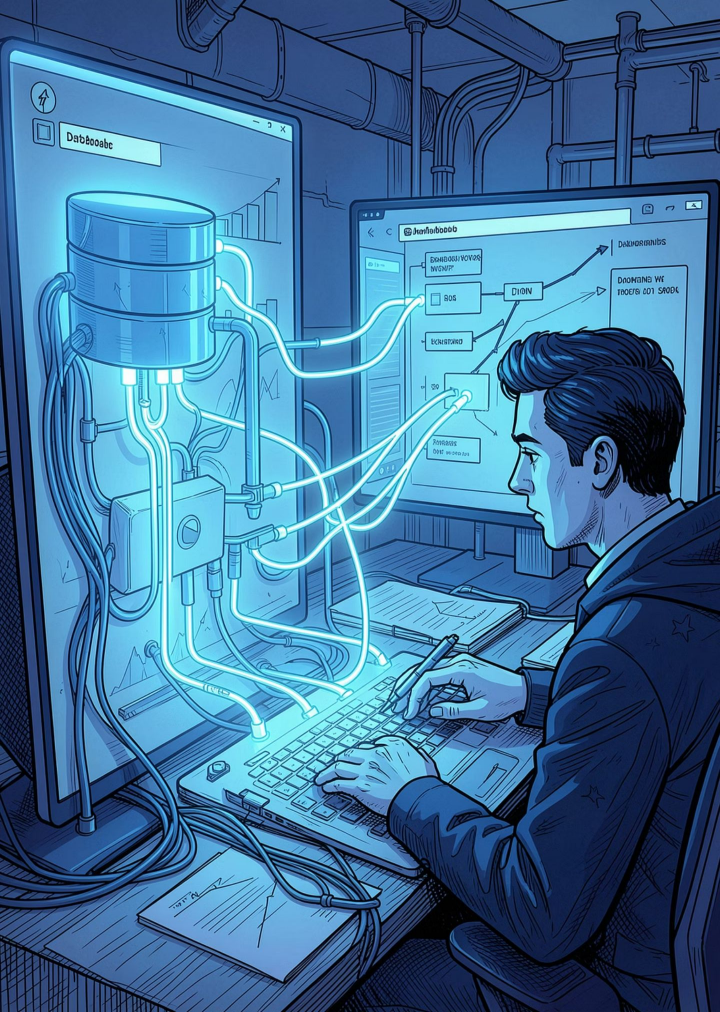
Poor Marketing Returns

Campaigns not converting at anticipated rates despite significant investment.

Key Performance Metrics

Four KPIs drive the entire analysis:

Metric	Definition	Purpose
Conversion Rate	% of visitors who purchase	Funnel effectiveness
Engagement Rate	Clicks, likes & comments	Content relevance
Average Order Value	Average spend per transaction	Revenue quality
Feedback Score	Average rating from reviews	Satisfaction level



Analytical Goals

Conversion Rates

1

Identify drop-off stages (Awareness → Consideration → Purchase) and recommend retargeting or checkout optimization.

Customer Engagement

2

Analyze views, clicks, and likes by content type (Social Media, Blog, Video) to guide content strategy.

Customer Feedback

3

Surface recurring themes in reviews to guide product improvements and service responses.

SQL Data Preparation

Raw data from the **MarketingAnalytics** database is cleaned and structured into analysis-ready tables:

dim_product

Products categorized as Low / Medium / High price tiers.

dim_customers

Customer records enriched with geographic data via LEFT JOIN.

fact_customer_reviews

Review text cleaned; double spaces removed before loading.

fact_engagement_data

Content types standardized; Views/Clicks split; Newsletters excluded.

fact_customer_journey

Duplicates removed via ROW_NUMBER(); missing durations imputed with daily average.



Sentiment Analysis — Python / VADER

Customer review sentiment is quantified using **NLTK's VADER**, suited to short, informal product reviews. Reviews are fetched from SQL Server via pyodbc. VADER produces a compound score in the range **[-1, +1]**; values above 0.05 indicate positive sentiment, below -0.05 indicate negative.



Enriched dataset exported as CSV with **SentimentScore**, **CategoryofSentiment**, and **SentimentBucket** columns for Power BI ingestion.



Positive

Score > 0.05 · 4–5 stars

Mixed Positive

Score > 0.05 · 3 stars

Neutral

Score -0.05 to 0.05

Mixed Negative

Score < -0.05 · 3 stars

Negative

Score < -0.05 · 1–2 stars

Power BI — Overview Dashboard

9.6%

Conversion Rate

Overall site conversion

9M

Social Views

Total social media views

2M

Clicks

Total clicks across campaigns

414K

Likes

Total content likes

3.69

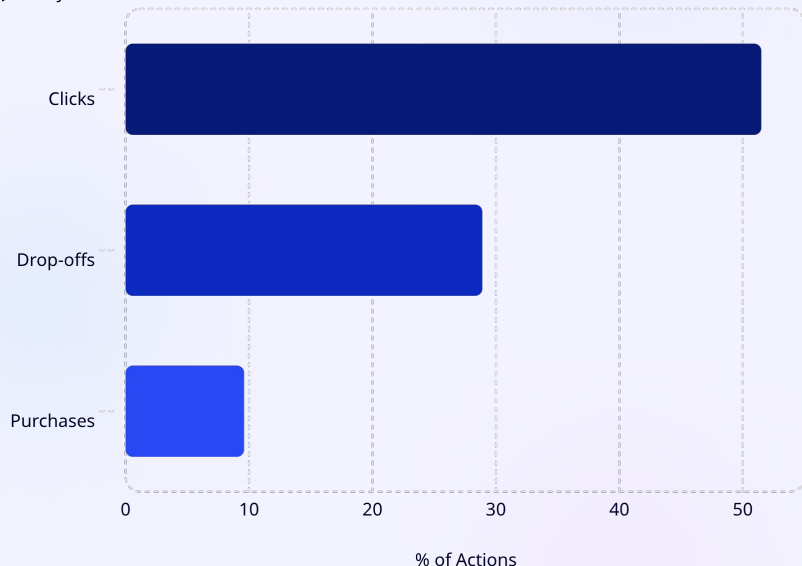
Avg Rating

Customer review average

Conversion rate peaks in **January (~18%)** and declines through the year. **Hockey Stick (~18%)** and **Ski Boots (~15%)** lead by product. Engagement metrics (Views, Clicks, Likes) show a general downward trend from January to December.

Conversion & Engagement Insights

Journey Action



Customer Journey Breakdown

Over half of all journey actions are Clicks, but **28.9% are Drop-offs** — a critical friction point. Only **9.6%** of actions result in a purchase.

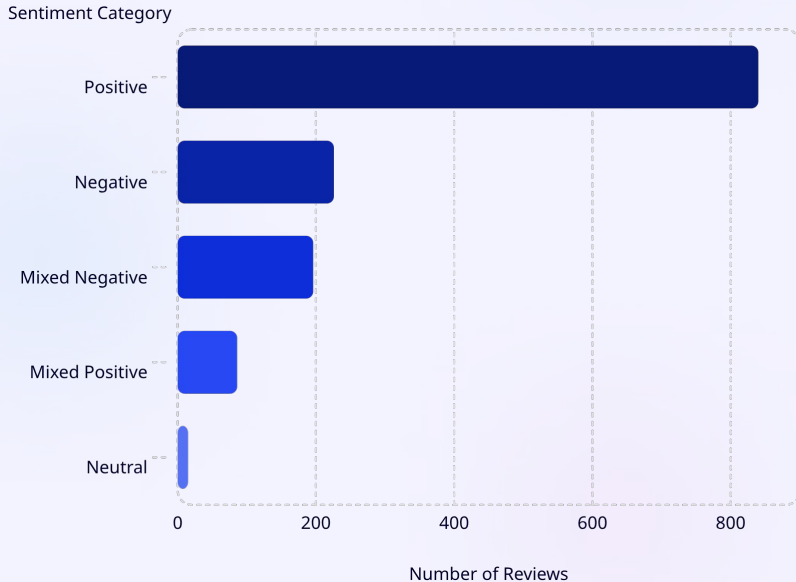
Content Type Reach

Blog, Social Media, and Video each hold **~34% share** of total views — an even split. Views trend downward across all months; top products by views include Swim Goggles, Tennis Racket, and Yoga Mat.



Newsletters are excluded from engagement KPIs as they operate on a different distribution model.

Customer Review Sentiment



1,363 Reviews Analyzed

61.6% of reviews are Positive (840), while **16.6%** are Negative (226). Ratings of 4 and 5 stars are most common, but 1-star and 2-star reviews pull the 3.69 average down.

Geographic Concentration

- **Spain** leads by country (18 customers), followed by Italy (12) and Germany (11)
- **Madrid** (18) and **Rome** (12) are top cities by customer count

Recommendations & Conclusion



Conversion Funnel

A/B test checkout CTAs; retarget consideration-stage visitors. Replicate Hockey Stick (15.5%) and Ski Boots (14.6%) patterns.



Content Strategy

Invest by cost-per-engagement, not volume. Investigate declining views – creative fatigue or budget shifts.



Customer Satisfaction

Address 226 negative reviews (16.6%); follow up on mixed/negative feedback to prevent churn.



Geographic Targeting

Prioritize Spain and Italy; explore expansion into underserved markets.

✓ **Baseline established:** 9.6% conversion rate, 3.69 avg rating. This end-to-end pipeline – SQL → Python → Power BI – enables measurable improvement.

